Fluent BioSciences

PIPseq™ T20 3′ Single Cell Capture and Lysis Kit v4.0

User Guide

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Introduction

The Fluent BioSciences PIPseq™ T20 3' Single Cell Capture and Lysis Kit uses Particle-templated Instant Partitions (PIPs) to capture individual cells with barcoded beads, lyse cells, and protect RNA in a stable state for up to 96 hours at 20°C. This represents the first part of the PIPseq T20 3′ Single Cell RNA workflow. The second part of the workflow beginning with mRNA isolation must be completed according to the FB0002130 PIPseq T20 Single Cell RNA Kit User Guide to produce sequencing-ready libraries. These libraries enable 3′ gene expression profiling of up to 80,000 individually captured cells or nuclei per kit.

Protocol Timing

Step	Duration	Stopping point
Reagent Preparation	~30 min	
General Cell Preparation Dependent on cell type	~1-1.5 hrs	
Cell capture and lysis	1 hr 40 min	20°C for up to 96 hours

Product Applications

The PIPseq T20 3′ Single Cell Capture and Lysis Kit is designed to complete the cell capture and lysis steps of the PIPseq T20 3′ Single Cell RNA workflow, pausing at a stopping point in which the sample is stable for up to 96 hours at 20°C. This stopping point enables users to ship the sample to a Fluent-authorized service provider to complete the gene expression library construction, sequencing and data analysis. This kit is configured with 4 individual reactions, in which up to 20,000 cells or nuclei can be profiled per reaction. It is recommended that a minimum of 2 samples are processed at a time.

PIPseq Platform Overview

The PIPseq platform consists of two reagent kits (Ambient and -80°C), a Consumable kit and a one-time Starter kit containing required equipment.

PIPseq T20 3′ Single Cell Capture and Lysis Kit Product Numbers

Product Name	Catalog Number	Storage
PIPseq T20 3′ Single Cell Capture and Lysis Ambient Kit v4.0	FBS-SCR-T20-A-4-V4-1	15°C to 30°C
PIPseq T20 3′ Single Cell Capture and Lysis -80°C Kit v4.0	FBS-SCR-T20-A-4-V4-2	-90°C to -75°C
PIPseq T20 3’ Single Cell Capture and Lysis Consumables Kit v4.0	FBS-SCR-T20-A-4-V4-3	15°C to 30°C
PIPseq T20 3′ Single Cell Starter Equipment Kit	FBS-SCR-T20-STKIT	15°C to 30°C



Warning: The items in the Starter kit below are highly recommended for the PIPseq 3′ Single Cell RNA workflow. The specific brand of plastic consumables have been validated to ensure stability of PIP emulsions. Substituting these materials may adversely affect performance.

PIPseq T20 3′ Single Cell Starter Equipment Kit

Component Name	Part Number	Units
PIPseq rotating vortex assembly for 1.5 mL tubes	FB0002809	1
PIPseq vortex mixer	FB0002373	1
PIPseq Dry bath with heated lid	FBS-SCR-PDB	1
PIPseq Dry bath 1.5 mL block	FB0002498	1
2.5 mm Allen Key Wrench	FB0001723	1
US Power Supplies for Vortex Mixer	FB0002353	1
US Power Supplies for PIPseq Dry bath	FB0002363	1
PIPseq 4-tube stand, blue, for 1.5 mL tubes	FB0002590	2
User Manual for PIPseq Dry bath	FB0002664	1
User manual for PIPseq Vortex Mixer	FB0002745	1
2.5 mm Allen key wrench, silver	FB0002666	1
PIPseq dry bath Stylus	FB0002667	1
Replacement tips for PIPseq dry bath Stylus	FB0002668	1
5x20 mm 5A 250V Fast blow fuse for PIPseq dry bath	FB0002669	1
Dry block removal tool, white	FB0002670	1

PIPseq T20 3′ Single Cell Capture and Lysis Consumables Kit

Component Name	Part Number	Units
3 mL syringe	FB0002625	4
G22 blunt bottom syringe needle	FB0002623	4
- Yellow 1.5 mL return Tube Holder	FB0002392	4

PIPseq T20 3′ Single Cell Capture and Lysis Reagent Kit Components

Component Name	Part Number	Kit	Storage	Units
T20 PIPs	FB0003914	PIPseq T20 3′ Single Cell Capture and Lysis -80°C Kit v4.0	**-90°C to -75°C**	4
Cell Suspension Buffer	FB0002440	PIPseq T20 3′ Single Cell Capture and Lysis -80°C Kit v4.0	**-90°C to -75°C**	4
Partitioning Reagent	FB0003123	PIPseq T20 3′ Single Cell Capture and Lysis Ambient Kit v4.0	15°C to 30°C	1
CLB3	FB0003910	PIPseq T20 3′ Single Cell Capture and Lysis Ambient Kit v4.0	15°C to 30°C	4

PIPseq Return Shipping Kit

Component Name	Part Number	Units
Return shipping kit	FB0004291	1
- Anti-static Reclosable Poly Bags 6” x 6”	FB0002642	1
- 10" x 12" 8 MI Reclosable Bags	FB0002644	1
- Bubble mailer 11" x 16"	FB0002646	1
- Adhesive Bubble wrap	FB0001836	1
- Cool Packs (5" x 6")	FB0004342	2

Reagent, Equipment, and Consumable Requirements

Required Third Party Reagents

Reagent	Supplier
Cell Counting Materials using laboratory′s preferred method (e.g. AO/PI)	Preferred Vendor
Nuclease-free water	Thermo Fisher; Cat # AM9937 or preferred laboratory supplier

Required Third Party Consumables

Item	Supplier
NOTE* Sterile, low retention tips are required for this protocol. Multiple suppliers may be used however only the below listed suppliers have been tested in this protocol	
Sterile Tips 20 µL, filtered, low retention	Rainin; Cat # 30389226 Filtrous; Cat # PTF-LS-0020 VWR;76322-528
Sterile Tips 200 µL, filtered, low retention and Sterile Tips 200 µL, filtered, low retention, wide-bore	Rainin; Cat # 30389240, 30389241 Filtrous; Cat # PTF-LS-0200 VWR; Cat #76322-150
Sterile Tips 1000 µL, filtered, low retention and Sterile Tips 1000 µL, filtered, low retention, wide-bore	Rainin; Cat # 30389213, 30389218 Filtrous; Cat # PTF-LS-1000 VWR; Cat #76322-154
15 mL centrifuge tubes, sterile	General laboratory supplier, recommend Corning # 430790

Required Third Party Equipment

Description	Supplier
Benchtop centrifuge with swinging bucket rotor, suitable for 5 mL, 15 mL tubes	General laboratory supplier, recommend ST40
Ice bucket or cold blocks, suitable for 0.2mL and 0.5mL PCR tubes and 1.5mL microcentrifuge tubes	General laboratory supplier
Hemocytometer or automated cell counter	General laboratory supplier
Micropipettes, 1µL-1000µL capabilities	General laboratory supplier
Benchtop centrifuge with swinging bucket rotor, suitable for 5 mL, 15 mL tubes	General laboratory supplier, recommend ST40
Benchtop microcentrifuge 2000xg, suitable for 1.5 mL tubes and 0.5 mL tubes. **NOTE* The microcentrifuge should decelerate before stopping in order to prevent beads from returning to the solution**	USA Scientific; Cat # 2631-0006

For some reagents and consumables, multiples may be required for pre-PCR and post-PCR use. Dedicated items and workspaces should be separated for pre-PCR and post-PCR processes to avoid carryover contamination.

Best practices

Notes for Working with RNA

Before executing this protocol, become familiar with working with RNA. The following section provides general guidelines for working with RNA, but the guidelines below are not all-inclusive.

- Due to the ubiquitous presence of RNases, RNA is susceptible to degradation. RNases are robust nucleases specific to RNA. Unlike DNases, they are not easily denatured or inactivated.
- The most common external RNase contamination comes from the skin. Therefore, any item that will be used for RNA work should be not handled without gloves to maintain nuclease-free surfaces. Wear a clean laboratory coat or gown to prevent shedding of skin or hair in and around the RNA work surface.
- Dust and bacteria are also sources of RNases. Keep surfaces and items free of dust and work with RNA away from bacterial processes.
- Refrigerator and freezer handles should be treated as RNase-free surfaces and should only be opened with gloved hands.
- Maintain reagents, pipettes, and a work space specifically for working with RNA. Wipe down pipettes and the working space regularly with a laboratory alcohol cleaning solution (70% alcohol). Periodically (once a week to once a month), clean surfaces with a 10% sodium hypochlorite solution. Alternatively, commercial solutions are available to inactivate RNases.

- Only use consumables and reagents that are nuclease-free and dedicated for RNA use. Reagents should be thoroughly thawed and aliquoted. In case of contamination, an aliquot can be discarded without compromising the entire reagent stock.

Cell Loading

This protocol describes addition of 40,000 cells into each PIPseq reaction, resulting in recovery of up to 20,000 cells and a multiplet rate of < 8%. The optimal input cell concentration is 4,000 cells per microliter. Alternative cell loading concentrations are allowed, as long as input volume is maintained, and increased cell loading will impact the observed multiplet rate according to the table below. Note that these data were obtained from species mixture experiments with mammalian cell lines and may not be representative of expected results with other sample types.

# Cells loaded	# Cells recovered	Multiplet rate
20,000	~10,000	4%
40,000	~20,000	7%
60,000	~30,000	13%

PIPseq T20 Protocol

The PIPseq protocol generally takes about 2 hours from cell addition to the emulsion-stable stopping point.

Reagent Preparation

Upon receiving the PIPseq T20 3' Single Cell Capture and Lysis kits, remove the -80°C box with the Cell Suspension Buffer (FB0002440), and PIP reactions (FB0003914) from the dry ice shipping container and store in a -80°C freezer.

Prior to each experiment, thaw one aliquot of Cell Suspension Buffer per cell preparation.

Shipment Preparation

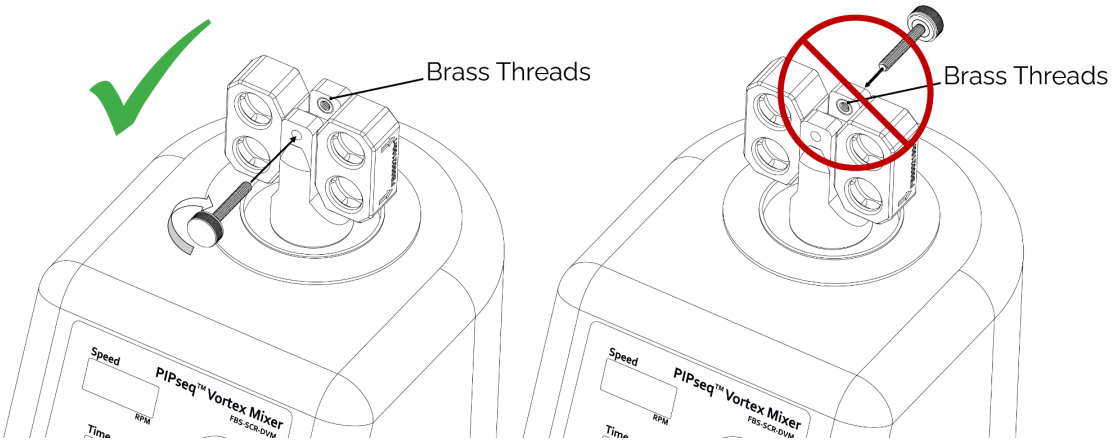
To ensure that the shipped emulsion will remain stable during shipment, **users must charge the provided gel packs 24 hours prior to shipment according to the following instructions**. For freeze protection during the winter, store the gel packs at room temperature which will enable them to remain in the liquid state. Do not freeze the gel packs. Use the gel packs while they are in the liquid state for shipping emulsions according to the Shipping Instructions at the end of this user guide.

For heat protection during the summer, place the gel packs into a refrigerator set to 2°C and let the gel packs charge for at least 24 hours. The cool packs will freeze and feel completely solid after this conditioning step. For quick charging, the gel packs can be placed in a -20°C freezer until they are solid. Then the gel packs should be transferred to a refrigerator after the charge so that the temperature stabilizes at 2°C before use. Ensure the gel pack is in the solid state prior to use for shipping emulsions according to the Shipping Instructions at the end of this user guide.

Equipment Preparation

IMPORTANT NOTE: Users will be required to alter the configuration of the rotating vortex adapter from horizontal to vertical during the PIPseq protocol. It is recommended that users practice altering the configuration of the adapter to ensure the process can be completed in 30 seconds prior to starting this protocol.

The vortex adapter head (yellow) is attached to the vortex base (gray) with a thumbscrew. The thumbscrew should be fastened from the front of the vortex mixer (see below). For further information on proper installation, review FB0003717 PIPseq Vortex Adapter Installation Guide.



The thumbscrew should be just tight enough such that the fixture does not rotate during vortexing, but not too tight such that changing from horizontal to vertical orientation takes significant effort. The tightness of the thumbscrew may need to be periodically adjusted. Users should ensure that the 1.5 mL tube block is affixed securely within the PIPseq dry bath prior to beginning this protocol. For every sample, assemble a syringe and a needle (provided).

Sample Preparation

The PIPseq single cell protocol requires a suspension of viable single cells or nuclei as input which may be derived from cell culture, dissociated tissues, cell sorting or other isolation methods. Users should minimize the presence of dead cells or aggregates to ensure the highest quality data. See the appropriate section for sample preparation instructions, for cells or nuclei.

Note: For sample types known to have high RNase content (e.g. pancreatic cells, monocytes), it is recommended that users consider adding RNase inhibitor into the PIP reaction (see Capture and Lysis, optional step).

Cell Preparation

This general cell preparation section describes a generic protocol for washing and resuspending cells starting from cryopreserved vials which is applicable to mammalian cell lines but may not be generalizable to all cell types (consult FB0001843 for more information on preparing cells for PIPseq). Preparation of single cell suspensions from tissues or fragile cell types may require additional dissociation or cell handling steps which are not described here. Users may substitute their own cell preparation protocol, however the final cell dilution step must be made using the Fluent Cell Suspension Buffer. Usage of wide-bore pipette tips is recommended to minimize cell damage.

1. Obtain a vial of the cryopreserved cells of interest from liquid nitrogen storage. If starting from fresh cells, skip to step 6.
2. Submerge the cryopreserved cell vial and an aliquot of Cell Suspension Buffer in a water bath set to 37°C ensuring that most of the tube is under the water line without being fully submerged.

3. After 1-1.5 minutes of thawing, check the cryopreserved cell vial. When there is only a moderate ice chunk remaining (60-70% thawed), remove the vial from the water bath. Decontaminate the outside of the vial with alcohol and move it into the biosafety cabinet. The remaining ice will thaw over the next 30-60 seconds at room temperature.
4. Use a **wide-bore** P1000 or 2mL serological pipette to transfer the cell suspension to a 15 mL conical tube. Gently pipette up and down to collect all cell contents from the tube.
5. **SLOWLY** add 9 mL of warmed thawing media to the cell suspension in the 15 mL conical tube and mix by inversion three to five times. The initial 2 mL should be added over **at least** 30 seconds with the remaining media added at an increasing pace.
6. Centrifuge cells at 200 x g for 5 min to pellet cells.

⚠ The cell pellet may form on the side or the bottom of the tube depending on the rotor type. Swinging bucket rotors are recommended for pelleting. Centrifugation speed depends on cell type and size. Generally, up to 300g x 5min can be used safely for smaller cell types. The goal is to form a loosely packed pellet with little to no cells in the supernatant. To this end, when working with new cell types it is important to examine the supernatant and monitor the difficulty of resuspension (such as pellet packed too tight) until you find the minimum speed and time required for optimal pelletting.

7. Aspirate as much of the supernatant as possible without disturbing the cell pellet.
8. ⚠ Add 1 mL pre-warmed Cell Suspension Buffer, gently mix 5 times with a P1000 **wide-bore** low retention pipette tip. Place the remaining Cell Suspension Buffer on ice to cool.
9. Centrifuge cells at 200 x g for 3 min to pellet cells. Aspirate as much of the supernatant as possible without disturbing the cell pellet.
10. ⚠ Using a **standard** bore low retention pipette tip, add 200-400 µL of cold Cell Suspension Buffer and gently mix 10-15 times until the cells are completely resuspended. If clumps are visible, increase the amount of force used during pipetting. The goal is to use the minimum force necessary to create a homogeneous cell suspension.

OPTIONAL STEPS: Users may choose to skip use of the tip strainer to minimize volume and cell concentration losses that can occur during filtration.

- ⚠ Using a **wide-bore** low retention pipette tip, withdraw 200 µL of the cell suspension and add a 40 micron FlowMi tip strainer onto the end of the tip.
- Gently dispense the cell suspension through the tip strainer into a fresh 1.5 mL tube. Do not "blow out" while pushing the cell suspension through the strainer as this will push unwanted debris through it and back into the cell suspension.

11. Determine cell concentration using the laboratory's preferred method (e.g. trypan blue staining on an automated cell counter).

⚠ Before proceeding, cells should be >90% viable. If cells have low viability, spin and count again to remove as much necrotic material as possible.

12. Prepare a cell suspension in Cell Suspension Buffer at a concentration of 4000 live cells per microliter using a **wide-bore** low retention pipette tip.
13. Once the target concentration is achieved, place the cells on ice.
14. Proceed to the Cell Capture and Lysis section.

Nuclei Preparation

Fluent recommends that users prepare nuclei suspensions using the PIPseq Nuclei Isolation Kit (FBS-SCR-NUC4) to ensure consistent, high quality nuclei across a diversity of tissues. See FB0003716 PIPseq Nuclei Isolation Kit User Guide for instructions.

IMPORTANT NOTE: The lysis temperature for nuclei is 66°C while the lysis temperature for cells is 37°C. At step 13 of Capture and Lysis, be sure to use the appropriate lysis temperature according to the sample type of interest.

Capture and Lysis

1. Users may choose to process 2-4 samples. Thaw the desired number of PIP tubes, one for each sample that will be processed.

   WORKFLOW TIP: Begin preheating the PIPseq Dry bath and lid to the appropriate temperature depending on sample input type (refer to table in step 13). Set the time to hold for this step as users will skip to the next step once they reach step 13 of this section.

2. Thaw the PIP tubes at room temperature for 15 minutes. After the PIP tubes have fully thawed, centrifuge them for ~5 seconds on a benchtop minifuge to remove air bubbles then place them on ice.

3. Mix the cell or nuclei suspension 10 times with a **wide-bore** P200 tip set to 80% of the cell suspension volume.

4. Add exactly 10 µL of prepared cells or nuclei (40,000 total) directly into the PIPs while avoiding the creation of air bubbles. If performing multiple reactions, add cells to all of them, no need to mix between additions.

   Optional Step: If working with sample types that have high RNase content users may add RNase Inhibitor into the cell:PIP mixture before moving onto step 5. Be advised that it is preferred that users prepare higher concentration cell input to ensure the total cell volume + RNAse inhibitor volume remains at 20 µL.

5. Mix the cell:PIP mixture 10 times using a **standard bore**, low-retention P200 tip at 180 µL stroke while moving the tip throughout the mixture starting from the bottom, moving up through the mixture, and returning to the bottom. Take care not to create bubbles, especially in the first six mixing strokes. For this step, we recommend using a P200 multichannel (in a staggered position).

   Troubleshooting: If foaming occurs, users may centrifuge the PIP tubes for ~ 5 seconds to remove the bubbles. Repeat the mixing in step 5 while avoiding excessive foaming.

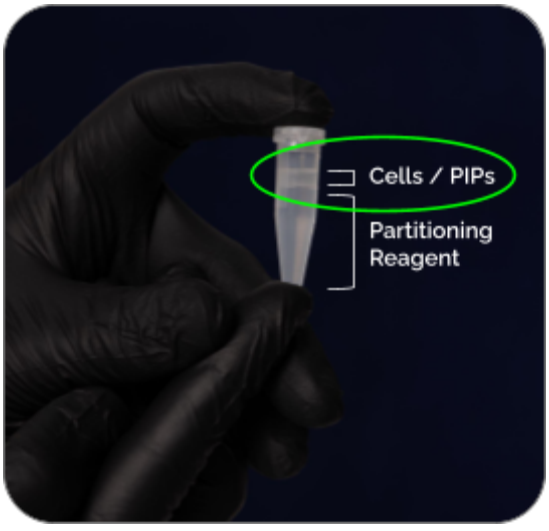


Figure 1. (circled in green) Cells and PIPs mixture floating on top of excess Partitioning Reagent

6. Add 1000 µL Partitioning Reagent, along the side wall of the tube, to the cell:PIP mixture (Figure 1) using a P1000 pipette. Do not use serological pipettes.

7. Ensure the tube caps are tightly sealed and place the tubes in the rotating vortex adapter in the horizontal configuration (Figure 2A). Ensure the tubes are fully depressed into the adapter. Vortex at 3000 RPM for 15 seconds.

8. Rotate the vortex adapter into the vertical configuration (Figure 2B). Vortex vertically at 3000 RPM for 2 minutes.



Figure 2. The rotating vortex adaptor shown in the horizontal configuration **(A)** and the vertical configuration **(B)**.

9. Remove the PIP tubes from the vortex adapter and place them into the 1.5 mL stand (FB0002590). Let emulsion stabilize for 30 seconds and then proceed to the next step. The reaction will first appear cloudy then the PIP emulsion will separate and two phases can be observed (Figure 3).

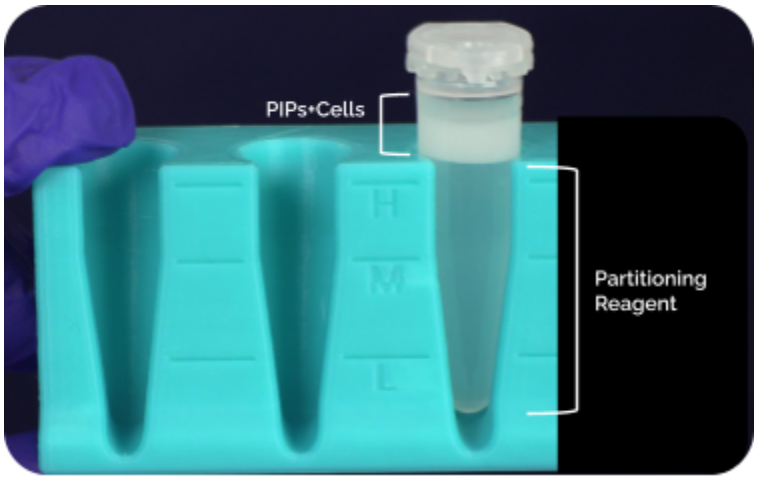


Figure 3. Emulsified PIPs and cells are seen floating in the top phase while the excess Partitioning Reagent in the bottom phase.

10. Slowly transfer Partitioning Reagent (bottom) out of each tube of PIPs using a 3 mL syringe attached with a G22 blunt tip syringe needle (Figure 4A) until the bottom of the PIPS+cells emulsion is aligned with the low ("L") volume marker of the stand (Figure 4C). Do this by slowly moving the tip through the emulsion to the bottom of the tube, wait for five seconds, and then aspirate out the bottom phase (Figure 4B). Be careful not to aspirate out any emulsion.

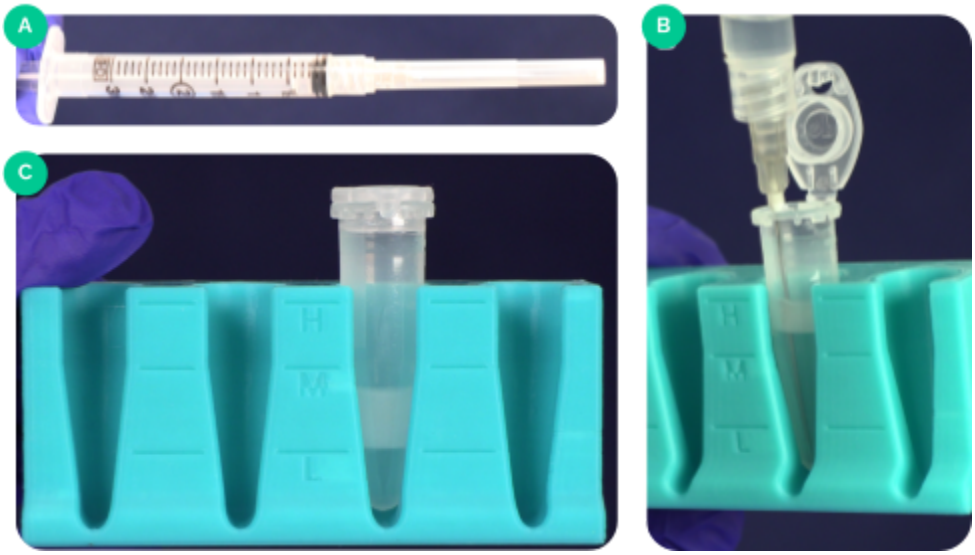


Figure 4. Use a 3 mL syringe with a G22 blunt bottom needle shown in **(A)** to remove excess Partitioning Reagent from the bottom phase as shown in **(B)**. Remove excess until the PIPs layer sits at the "Low" or "L" bar of the blue tube rack as shown in **(C)**.

⚠ Tips for pipetting through an emulsion: Be cognizant of emulsion sticking tightly to the end of the needle. It is usually helpful to wipe any tip that has passed through an emulsion on the side of the tube on the way out.

11. Prepare one Chemical Lysis Emulsion per reaction by adding 630 µL of Partitioning Reagent to the provided CLB3 (FB0003910) tube containing chemical lysis buffer. Vortex for 10 seconds to generate the emulsion and immediately use a P1000 low-retention tip to add the whole volume of the Chemical Lysis Emulsion (~840 µL) on top of the PIP emulsion in the PIP tubes. Use one tube of Chemical Lysis Emulsion per sample.

12. Mix the combined emulsions gently by inversion 10 times.

⚠ **Ensure the volume of the emulsion is below the top of the thermal block to ensure consistent heating.**

13. Verify the PIPseq Dry bath (Figure 5) is preheated to the appropriate temperature according to the sample type in use (see table below) with the lid temperature set to +5°C for cells or 105°C for nuclei. Refer to the *PIPseq Dry Bath Lid Settings and Control section* of this user guide for instructions on how to program the desired heated lid setting. After at least 10 minutes of preheating is complete, insert samples and then skip to the next step to begin the lysis incubation as indicated in the table below.



Figure 5. The PIPseq™ Dry bath to be used for the lysis incubation step.

Cells		
Lid Temp	**+5 °C**	
Preheating step	25 °C	hold (“0000”)
After preheating complete and samples inserted, skip to next step		
	25 °C	15 min
	37 °C	45 min
	25 °C	10 min
	20 °C	hold

Nuclei		
Lid Temp	**105 °C**	
Preheating step	66 °C	hold (“0000”)
After preheating complete and samples inserted, skip to next step		
	66 °C	45 min
	25 °C	10 min
	20 °C	hold



14. After incubation is complete, the samples can be held at 20°C for up to 96 hours (**Do not freeze**!) before proceeding to the second part of the protocol following the PIPseq T20 Single Cell RNA Kit User Guide starting at mRNA Isolation. If shipping the samples out for processing at Fluent-certified service facilities, proceed to Shipping Instructions below. We recommend shipping samples immediately, so that samples will reach their destination within 96 hrs after lysis. Note: Some condensation may appear on the tubes, which is not a concern.

Shipping Instructions

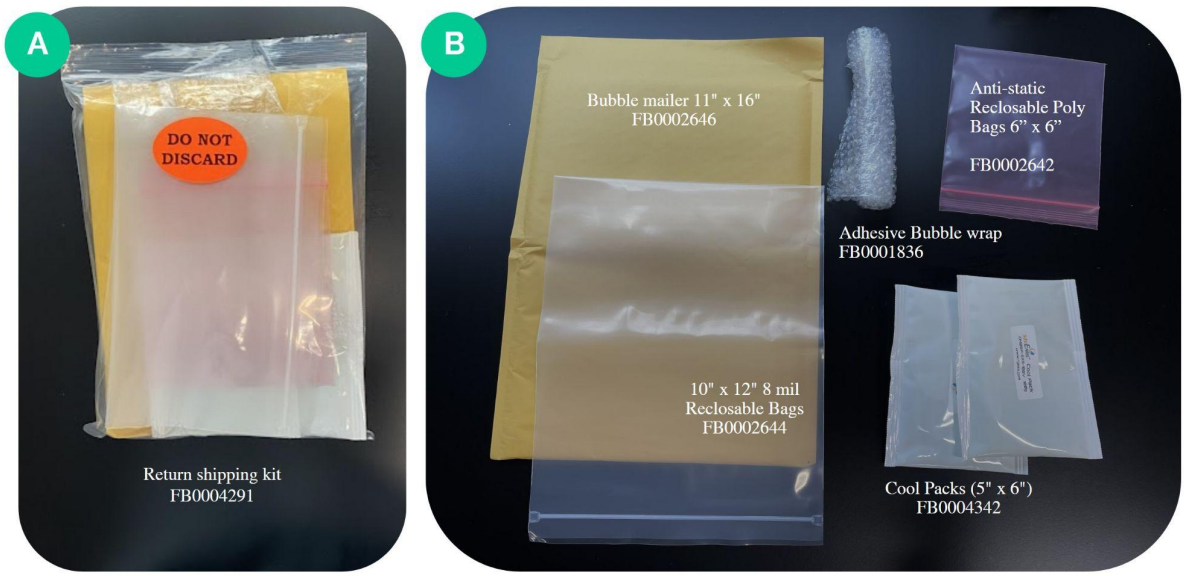


Figure 6. Return shipping kit (A) with the individual components separately labeled (B).

1. Remove samples from the PIPseq Dry Bath when ready to ship and add sufficient volume (~500 µL) of Partitioning Reagent to each sample such that the total volume is just below the top lip of the tube. This serves to minimize headspace in the tube and protect the emulsion layer during shipping.
2. Attach a Fluent barcode bib-tag to each sample (Figure 6).



Figure 7. T20 sample tube with Fluent barcode bib-tag attached.

3. Snap PIP tube into Shipping Secure Tube Holder.
    a. Note which barcode is associated with each of your samples:
        i. Barcode 1:_____ Sample ID: _____
        ii. Barcode 2:_____ Sample ID: _____
        iii. Barcode 3:_____ Sample ID: _____
        iv. Barcode 4:_____ Sample ID: _____

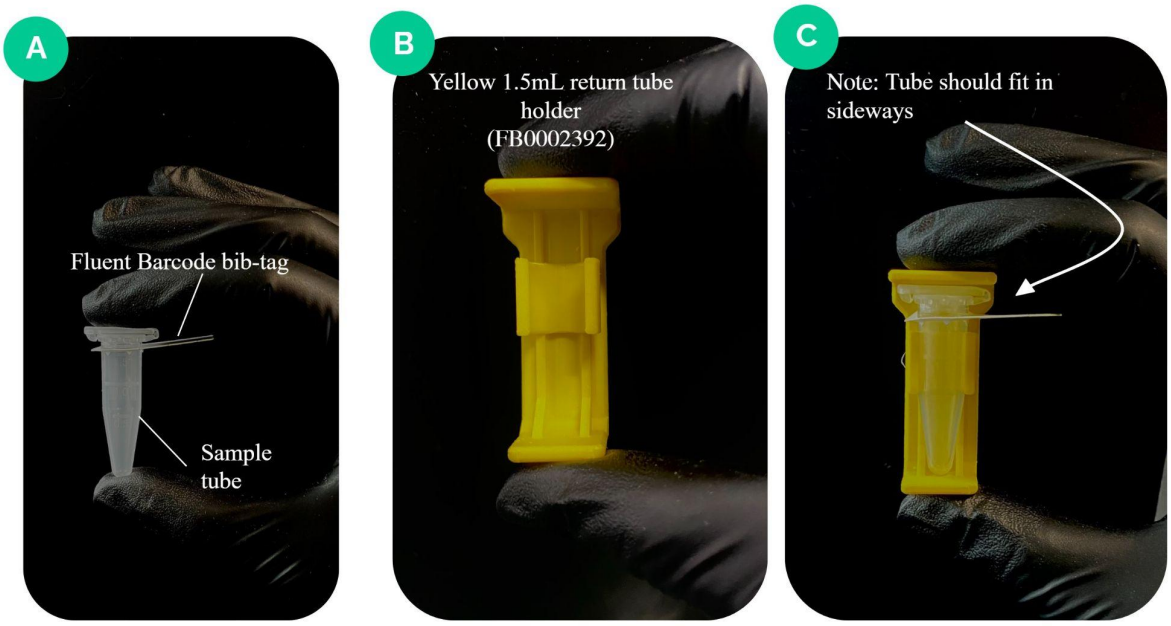


Figure 8. T20 sample tube with Fluent barcode bib-tag attached (A). Yellow shipping holder for securing sample (B). Secured sample with the Barcode bib-tag properly affixed (tube secured sideways) (C).

4. Ensure samples are secure inside the shipping holder(s) and place them in the small anti-static plastic bag. Expel the excess air as you close the bag (Figure 9A).
5. Fold the anti-static bag with the samples and place it inside the medium plastic bag, sandwiched between two cool packs (Figure 9B).



Figure 9. Place the sample secured in the return tube holder inside the anti-static bag (A). Fold the bag and expel the air prior to sandwiching between two ice packs inside a medium plastic bag (B).

6. Seal the medium plastic bag, wrap with bubble wrap and seal tightly (Figure 10).



Figure 10. Seal the medium plastic bag, ensuring the samples are still between the cool packs (A). Wrap the bubble wrap around the medium plastic bag (B) and seal tightly (C).

7. Place the sealed medium plastic bag wrapped in bubble wrap inside the yellow bubble mailer envelope as shown in Figure 11A. Seal the yellow bubble mailer envelope and affix the return label to the outside of the bubble mailer (Figure 11B).

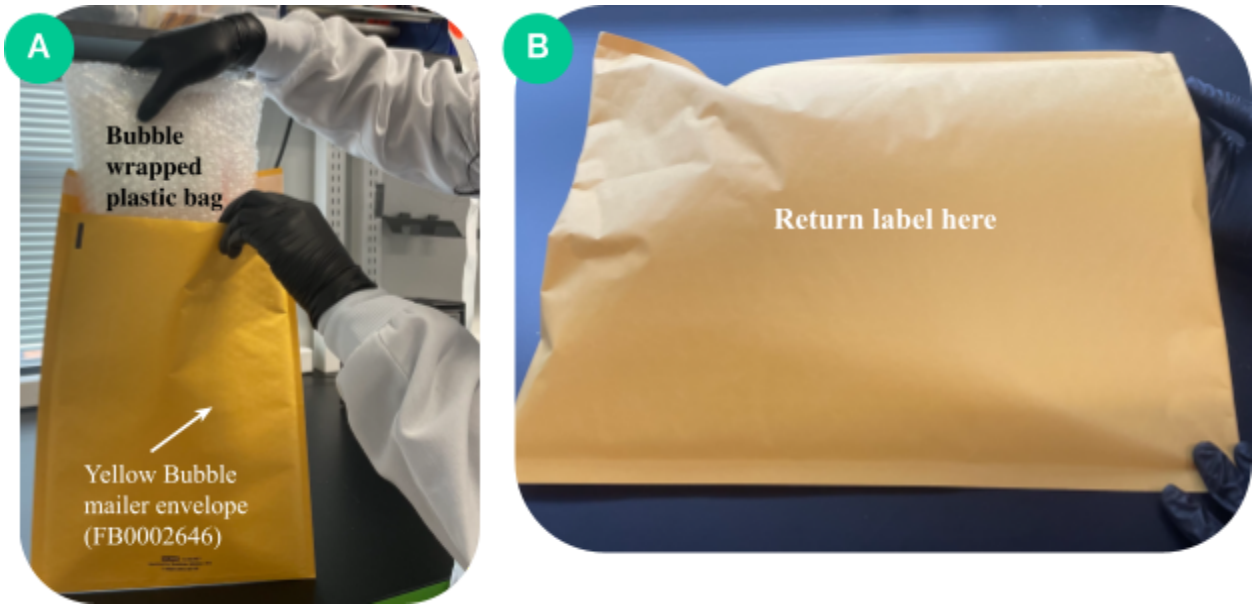


Figure 11. Place the bubble-wrapped plastic bag in the yellow bubble mailer envelope (A). Add the return label in the center of the envelope (B).

8. Ship the package and ensure that samples will reach their destination within 96 hrs after lysis.

Note: Please note that if you are sending samples to Fluent-certified service providers for downstream processing, the sample shipping deadline is Wednesday every week for Thursday delivery. For more information contact Fluent Services Support at services@fluentbio.com. We recommend that customers plan their shipments ahead of time, based on the daily package pickup/dropoff cut-off at your local FedEx facility, to avoid any delays due to federal holidays to ensure samples are received at Fluent in a timely manner.

Abbreviations

PIPs	Particle-templated Instant Partitions
RNA	Ribonucleic Acid
cDNA	Complementary DNA
RPM	Revolutions Per Minute

Document Revision Summary

Doc ID: **FB0002569**
Revision: **8.2**
Revision date: June 2023

General Changes:

- Updated pictures in shipping section to include gel packs and new shipping configuration

Specific Changes:

- Change to gel shipping packs enabling 96 hours of emulsion-stability

Legal Notices

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